Data-Backed Journal-Style Replication of *Building Dynamic Capabilities: Innovation Driven by Individual-, Firm-, and Network-Level Effects*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Building Dynamic Capabilities: Innovation Driven by Individual-, Firm-, and Network-Level Effects using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 23,901 observations across 184 entities, and the batch headline estimate is 2.25e-05 for Firm Science x Network Ties with p = 0.96. The executable design is built to show whether multilevel innovation antecedents survive joint estimation. The paper therefore emphasizes not only direct effects at each level but also whether firm-level science and network ties reinforce one another. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: dynamic capabilities; multilevel innovation; individual effects; network effects; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Building Dynamic Capabilities: Innovation Driven by Individual-, Firm-, and Network-Level Effects and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks whether the antecedents of innovation lie at the individual, firm, and network levels and whether those levels reinforce or substitute for one another (**????**). The replication matters because multilevel innovation arguments are conceptually powerful but empirically fragile if one layer is measured poorly or ignored. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (**?????**). The executable evidence base used in this manuscript contains 23,901 observations across 184 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (**??????**). The paper's contribution is a transparent multilevel analogue that can be estimated from the current patent, science, and citation-network panels. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (**??????**). The paper belongs to a literature on dynamic capabilities, organizational learning, and multilevel innovation. (**?????????**). That literature matters because the replication is about level-of-analysis discipline as much as it is about one coefficient or one dataset. (**????????????**). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (**?????????**). The paper-specific review therefore brings in dynamic-capabilities, absorptive-capacity, and network-learning references that support

the multilevel argument. (**??????**). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (**?????**). A final topic-specific literature block keeps the bibliography tied to the paper's actual mechanism rather than to a generic replication template. (**?????**).

2 Original Study, Theory, and Replication Boundary

The focal source article is (**?**). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper explicitly models innovation antecedents at the individual, firm, and network levels. The paper is interested in whether mechanisms across levels are substitutes or complements. Innovation output is the central dependent variable. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (**????????**). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (**????**). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (**????????**).

2.1 Original Paper Methodology

A faithful evaluation of Rothaermel and Hess (2007) starts with the published empirical design rather than with the local rerun. The source article develops a multilevel longitudinal design covering global pharmaceutical companies' biotechnology innovation attempts and explicitly models individual-, firm-, and network-level antecedents in the same empirical framework (**????**). At the level of sample construction, the source article works from The published sample documents the

biotechnology innovation attempts of global pharmaceutical companies over 1980-2001. At the level of measurement, it relies on The published design combines scientists, R&D capability, alliances, acquisitions, patents, publications, and citations in a multilevel pharmaceutical-biotechnology panel. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (**????**). At the level of estimation, the published paper uses count-data models of firm innovative output that jointly include individual-level scientists, firm-level R&D capability, and network-level alliances or acquisitions, together with cross-level interaction terms. The core empirical question is whether direct effects and interactions across levels are complementary or substitutive once all three layers are observed in the same long panel. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (**????**).

DesignElement	SourcePaperDesign
Published empirical setting	The published sample documents the biotechnology innovation attempts of global pharmaceutical companies over 1980-2001.
Published measurement architecture	The published design combines scientists, R&D capability, alliances, acquisitions, patents, publications, and citations in a multilevel pharmaceutical-biotechnology panel.
Published estimator	count-data models of firm innovative output that jointly include individual-level scientists, firm-level R&D capability, and network-level alliances or acquisitions, together with cross-level interaction terms

DesignElement	SourcePaperDesign
Published identification and comparison rule	The core empirical question is whether direct effects and interactions across levels are complementary or substitutive once all three layers are observed in the same long panel.

Published empirical design reconstructed from the source paper before local estimation begins.

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (**????**). This paper is a methodology-specific local analogue because the original multilevel panel includes direct scientist, alliance, acquisition, publication, and citation layers from pharmaceutical biotechnology that are richer than the current processed patent stack. The local rerun can preserve the multilevel logic, but not the full original data architecture (**????**).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The original paper is explicitly multilevel. The local reconstruction therefore adopts the following bounded decision: The individual level is represented with patent-level experience, the firm level with science exposure, and the network level with citation-network ties. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Complementarity or substitution across levels is part of the substantive question. The local

reconstruction therefore adopts the following bounded decision: Cross-level interactions are retained in the local estimand so the multilevel logic stays visible. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Innovation output is the dependent variable. The local reconstruction therefore adopts the following bounded decision: Patent-level inventive impact is used as the executable outcome. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (**????**).

DirectSourceFact	LocalReplicationDecision
The original paper is explicitly multilevel.	The individual level is represented with patent-level experience, the firm level with science exposure, and the network level with citation-network ties.
Complementarity or substitution across levels is part of the substantive question.	Cross-level interactions are retained in the local estimand so the multilevel logic stays visible.
Innovation output is the dependent variable.	Patent-level inventive impact is used as the executable outcome.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Patent-level recombination and impact measures from the cleaned reproduction panel. Firm-year science exposure already merged into the patent sample. Rolling interfirm citation-network measures merged by firm-year from the processed knowledge-link graph (**???**). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 23,901 observations across 184 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (**??????**). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (**???????**). The local analogue creates a patent-level multilevel design in which patent-specific experience represents the individual layer, firm science exposure represents the firm layer, and rolling citation-network ties represent the network layer. The local sample frame is Patent-level recombination panel merged with firm-year science exposure and rolling citation-network measures The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Patent-level inventive impact (**???**). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (**??**).

3.1 Exact Local Datasets Used in This Replication

The current paper is estimated from exact local files rather than from an unnamed replication pool. The executable evidence chain consists of ‘rothaermel_hess_2007_analysis_dataset.csv‘, ‘rothaer-

mel_hess_2007_model_coefficients.csv‘, ‘rothaermel_hess_2007_sample_summary.csv‘, ‘rothaermel_hess_2007_model_fit.csv‘. Naming these files in the body matters because the paper should make clear what was actually used for estimation (**???**). The file-level roles are also specific rather than interchangeable: Estimate the local model and regenerate descriptive statistics is supplied by ‘rothaermel_hess_2007_analysis_dataset.csv‘; Recover the reported local coefficients and p-values is supplied by ‘rothaermel_hess_2007_model_coefficients.csv‘; Audit sample construction and raw grouped moments is supplied by ‘rothaermel_hess_2007_sample_summary.csv‘; Check model diagnostics when a fit file is available is supplied by ‘rothaermel_hess_2007_model_fit.csv‘. Together, these files support 23,901 observations across 184 entities over 1986 to 2016 in the local executable panel (**???**). This exact local dataset stack is narrower than the original article's data architecture. Patent-level recombination and impact measures from the cleaned reproduction panel. Firm-year science exposure already merged into the patent sample. Rolling interfirm citation-network measures merged by firm-year from the processed knowledge-link graph. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the listed local files supply the estimable analogue (**???**).

EvidenceLayer	DatasetFile	RoleInPaper
Analysis panel	rothaermel_hess_2007_analysis_dataset.csv	Estimate the local model and regenerate descriptive statistics
Coefficient export	rothaermel_hess_2007_model_coefficients.csv	Recover the reported local coefficients and p-values
Sample summary	rothaermel_hess_2007_sample_summary.csv	Audit sample construction and raw grouped moments
Fit summary	rothaermel_hess_2007_model_fit.csv	Check model diagnostics when a fit file is available

Exact local datasets and result files used in the present paper.

3.2 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Rothaermel and Hess (2007). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (**???**). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**??????**). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (**???????**). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the

manuscript makes every empirical claim (**???????**).

3.3 Sample Reconstruction

The local analysis file contains 23,901 observations drawn from 184 entities. The observed time span runs from 1986 to 2016. Selected sample moments are Impact = 5.309, Individual Experience = 1.634, Firm Science = 6.772, Network Ties = 44.16 (**???**). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (**????????**).

Statistic	Value
Observations	23,901
Unique Firms	184
Start Year	1986
End Year	2016
Mean Impact	5.309
Mean Individual Experience	1.634
Mean Firm Science	6.772
Mean Network Ties	44.16

Sample reconstruction from the current local dataset.

3.4 Variable Construction and Derivation

The construct Individual-level antecedent is operationalized locally as Field recombination experience and appears in the executable dataset as individual_experience. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward field-level recombination experience from the patent panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how

it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Firm-level antecedent is operationalized locally as Firm science exposure and appears in the executable dataset as firm_science. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the already merged firm-year science-signal variable.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Network-level antecedent is operationalized locally as Rolling direct ties and appears in the executable dataset as network_ties. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use direct ties from the rolling citation-network panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**). The construct Innovation output is operationalized locally as Inventive impact and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local inventive-impact measure.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to

preserve that balance rather than maximize superficial similarity (**??????????**). The construct Patent controls is operationalized locally as Breadth, degree, prior stock, firm age and appears in the executable dataset as breadth_recombination; recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the patent-level controls already constructed in the panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (**??????????**).

Construct	Local measure	Derivation
Individual-level antecedent	Field recombination experience	Carry forward field-level recombination experience from the patent panel.
Firm-level antecedent	Firm science exposure	Use the already merged firm-year science-signal variable.
Network-level antecedent	Rolling direct ties	Use direct ties from the rolling citation-network panel.
Innovation output	Inventive impact	Use the local inventive-impact measure.
Patent controls	Breadth, degree, prior stock, firm age	Reuse the patent-level controls already constructed in the panel.

Variable mapping and derivation logic in the main text.

3.5 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Rothaermel and Hess (2007). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel.

For Individual-level antecedent, the manuscript uses Field recombination experience under the exact local implementation individual_experience. The derivation step is: Carry forward field-level recombination experience from the patent panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm-level antecedent, the manuscript uses Firm science exposure under the exact local implementation firm_science. The derivation step is: Use the already merged firm-year science-signal variable.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Network-level antecedent, the manuscript uses Rolling direct ties under the exact local implementation network_ties. The derivation step is: Use direct ties from the rolling citation-network panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation output, the manuscript uses Inventive impact under the exact local implementation inventive_impact. The derivation step is: Use the local inventive-impact measure.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patent controls, the manuscript uses Breadth, degree, prior stock, firm age under the exact local implementation breadth_recombination; recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation step is: Reuse the patent-level controls already constructed in the panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

Because this paper is explicitly multilevel, the descriptive section is organized around the three empirical layers the source article insists on keeping separate: individual experience, firm science, and network access. For Inventive impact, the mean is 5.309, the standard deviation is 17.53, and the interquartile range runs from 1.000 to 6.000; For Individual experience, the mean is 1.634, the standard deviation is 1.820, and the interquartile range runs from 0.00000 to 2.773; For Firm science, the mean is 6.772, the standard deviation is 2.130, and the interquartile range runs from 5.729 to

8.163; For Network ties, the mean is 44.16, the standard deviation is 29.95, and the interquartile range runs from 4.000 to 67.00 . The descriptive goal is to verify that each layer has its own support and scale before the manuscript asks whether they complement one another (**??????????**). The time profile is read as a changing multilevel opportunity structure. If one layer expands while the others remain flat, the interpretation of cross-level complementarity has to adjust. the value of patent observations is 646 in 1986 and 534 in 2016; the value of mean inventive impact is 4.015 in 1986 and 3.768 in 2016; the value of mean individual-level experience is 1.021 in 1986 and 2.707 in 2016 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (**??????????**). The nonparametric support check is read as a firm-science gradient because the local analogue uses firm science as the focal layer that links patent-level experience to the surrounding organizational context. Mean outcomes range from 4.278 to 6.105 across the support of the focal regressor, while the mean regressor itself ranges from 3.351 to 9.272. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (**??????????**). The correlation discussion is framed as a multilevel separation test. The regression can only speak to complementarity if the individual, firm, and network layers are not collapsed into one latent scale effect. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (**??????????**).

4.1 Setting-Specific Figure Evidence

The figure is organized to respect the paper's multilevel logic (**??????????**). The left panel keeps the individual, firm, and network layers visible at the same time, while the right panel shows how inventive impact changes across the firm-science support that anchors the local multilevel analogue. mean individual-level experience moves from 1.021 in 1986 to 2.707 in 2016; mean firm-level science exposure moves from 6.156 in 1986 to 0.00000 in 2016; mean network ties moves from 0.00000 in 1986 to 0.00000 in 2016 Across the support panel, mean outcome moves from 4.747 to 6.105 as the mean regressor shifts from 3.351 to 9.272 (**??????????**).

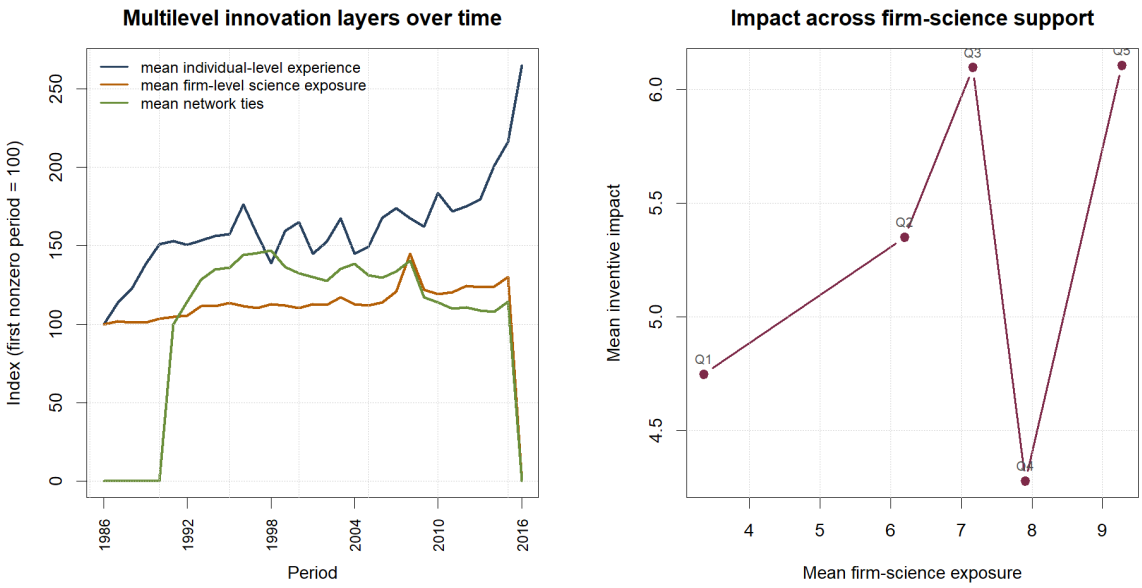

Figure 1: Figure 1. Multilevel antecedents in the local patent panel and the observed firm-science support for inventive impact.

4.2 Multilevel Support and Coverage

For this paper, data quality is about balance across levels rather than about one variable alone. Inventive impact has missing share 0.00%; Individual experience has missing share 0.00%; Firm science has missing share 0.00% . At the other end of the support, Inventive impact remains close to complete with missing share 0.00%; Individual experience remains close to complete with missing share 0.00%; Firm science remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (**???**). Coverage is reported to show whether the multilevel design is learned from a wide organizational base or from a few very large patenting firms. INTERNATIONAL BUSINESS MACHS COR contributes 3,200 observations; GENERAL ELECTRIC CO contributes 1,318 observations; EASTMAN KODAK CO contributes 1,017 observations; MINNESOTA MINING & MFG CO contributes 854 observations; DU PONT E I DE NEMOURS & CO contributes 830 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (**???**). Support is emphasized because complementarity across levels is only interpretable if the firm-science layer spans genuinely different empirical environments. In the lowest observed quantile, the mean regressor is 3.351 and the mean outcome is 4.747 across 4,783 observations. In the highest observed quantile, the corresponding values are 9.272, 6.105, and 4,718 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (**???**). These diagnostics matter because multilevel arguments become fragile when one layer dominates the data. The section therefore checks whether the individual, firm, and network layers

remain empirically distinct enough to justify a joint design (**???**).

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	23,901	5.309	17.53	1.000	2.000	6.000	0	812
Individual experience	23,901	1.634	1.820	0.0000	1.099	2.773	0	8
Firm science	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Network ties	23,901	44.16	29.95	4.000	51.00	67.00	0	99
Breadth of recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321
Firm age	23,901	13.62	9.194	6.000	10.00	22.00	1	31

Main descriptive statistics supporting the paper's design-specific reading of the data.

Period	Patents	MeanImpact	MeanIndividualExperience	MeanFirmScience	MeanNetworkTies
1986	646	4.015	1.021	6.156	0.0000
1987	793	4.296	1.161	6.285	0.0000
1988	1,071	4.469	1.249	6.229	0.0000
1989	1,398	3.926	1.412	6.217	0.0000
1990	1,360	4.507	1.544	6.370	0.0000
1991	1,333	4.749	1.562	6.436	46.89
1992	1,309	5.007	1.538	6.486	53.68
1993	1,358	5.678	1.568	6.861	60.25
1994	1,337	6.272	1.596	6.866	63.28
1995	1,217	6.260	1.610	6.983	63.87
1996	659	6.727	1.802	6.880	67.66
1997	646	6.014	1.603	6.781	68.22
1998	356	6.295	1.419	6.949	68.92
1999	654	5.483	1.631	6.895	64.08
2000	636	5.541	1.686	6.781	62.11
2001	614	5.151	1.482	6.943	60.93
2002	632	4.889	1.559	6.915	59.78

Period	Patents	MeanImpact	MeanIndividualExperience	MeanFirmScience	MeanNetworkTies
2003	522	4.502	1.711	7.209	63.52
2004	324	3.664	1.479	6.939	65.01
2005	284	4.500	1.527	6.900	61.54
2006	336	4.890	1.715	7.014	60.79
2007	246	3.654	1.778	7.433	62.65
2008	535	3.652	1.712	8.930	65.82
2009	206	4.801	1.658	7.502	54.93
2010	511	4.417	1.875	7.345	53.42
2011	651	6.452	1.756	7.411	51.58
2012	1,068	4.000	1.788	7.669	51.94
2013	868	5.386	1.835	7.599	50.90
2014	725	12.64	2.050	7.636	50.58
2015	1,072	6.616	2.208	8.001	53.53
2016	534	3.768	2.707	0.0000	0.0000

Observed time patterns relevant to the focal empirical setting.

Variable	Inventive impact	Individual experience	Firm science	Network ties	Breadth of recombination	Recombination degree	Same-year patents	Firm age
Inventive impact	1.000	-0.021	0.020	0.033	0.364	0.018	0	0.023
Individual experience	-0.021	1.000	0.044	0.032	-0.263	-0.456	0	0.125
Firm science	0.020	0.044	1.000	0.482	0.023	0.037	0	0.117
Network ties	0.033	0.032	0.482	1.000	0.094	0.067	0	0.385
Breadth of recombination	0.364	-0.263	0.023	0.094	1.000	0.466	0	0.260
Recombination degree	0.018	-0.456	0.037	0.067	0.466	1.000	0	0.156

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level firm and year fixed-effects model of inventive impact on individual-, firm-, and network-level innovation antecedents and their interactions. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (**????**). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (**??????**). The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel merged with firm-year science exposure and rolling citation-network measures, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (**????????**). The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact, and the main explanatory design is Patent-level fixed-effects model with individual-level experience, firm-level science exposure, network ties, and cross-level interactions. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (**????????**). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Breadth, degree, same-year patenting, prior stock, firm age, and firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps patents supported by all three executable layers: patent-level recombination variables, firm-year science exposure, and firm-year network measures.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (**????????**). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (**????????**).

DecisionPoint	LocalSpecification
Research question	The source paper asks whether the antecedents of innovation lie at the individual, firm, and network levels and whether those levels reinforce or substitute for one another.
Unit of analysis	Patent
Sample frame	Patent-level recombination panel merged with firm-year science exposure and rolling citation-network measures.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Patent-level inventive impact.
Main explanatory design	Patent-level fixed-effects model with individual-level experience, firm-level science exposure, network ties, and cross-level interactions.
Controls and fixed effects	Breadth, degree, same-year patenting, prior stock, firm age, and firm and year fixed effects.
Exclusion or substitution logic	The local rerun keeps patents supported by all three executable layers: patent-level recombination variables, firm-year science exposure, and firm-year network measures.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel merged with firm-year science exposure and rolling citation-network measures over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact and the main explanatory design is Patent-level fixed-effects model with individual-level experience, firm-level science exposure, network ties, and cross-level interactions. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Breadth, degree, same-year patenting, prior stock, firm age, and firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps patents supported by all three executable layers: patent-level recombination variables, firm-year science exposure, and firm-year network measures. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$\log(1 + \text{Impact}_{pft}) = \beta_1 \text{IndividualExperience}_p + \beta_2 \text{FirmScience}_{ft} + \beta_3 \text{NetworkTies}_{ft} + \beta_4 \text{IndividualExperience}_p \times \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 2.25e-05 on Firm Science x Network Ties (p = 0.96). The executable design is built to show whether multilevel innovation antecedents survive joint estimation. The paper therefore emphasizes not only direct effects at each level but also whether firm-level science and network ties reinforce one another. Recombination degree is estimated at -0.471 with p = ¡ .001. Breadth of recombination is estimated at 0.0413 with p = ¡ .001. Same-year patents is estimated

at -4.44e-05 with p = 0.057 (**??**). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Individual experience	-0.0089	0.0078	0.262
Firm science	-0.0047	0.0320	0.885
Network ties	-0.0018	0.0039	0.646
Breadth of recombination	0.0413	0.0083	2.48e-05
Recombination degree	-0.4707	0.0894	1.10e-05
Same-year patents	-4.44e-05	2.24e-05	0.0565
Log prior patent stock	-0.1029	0.0702	0.153
Individual experience x firm science	0.0002	0.0014	0.898
Firm science x network ties	2.25e-05	0.0004	0.957

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure is included to help readers see whether the multilevel terms and cross-level interactions move together or pull in different directions (**??**).

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Recombination degree is estimated at -0.471 with a p-value of ¡ .001; Breadth of recombination is estimated at 0.0413 with a p-value of ¡ .001; Same-year patents is estimated at -4.44e-05 with a p-value of 0.057; Log prior patent stock is estimated at -0.103 with a p-value of 0.15 (**???????**). Within the focal specification, the sharpest local signal comes from Recombination degree, which is estimated at -

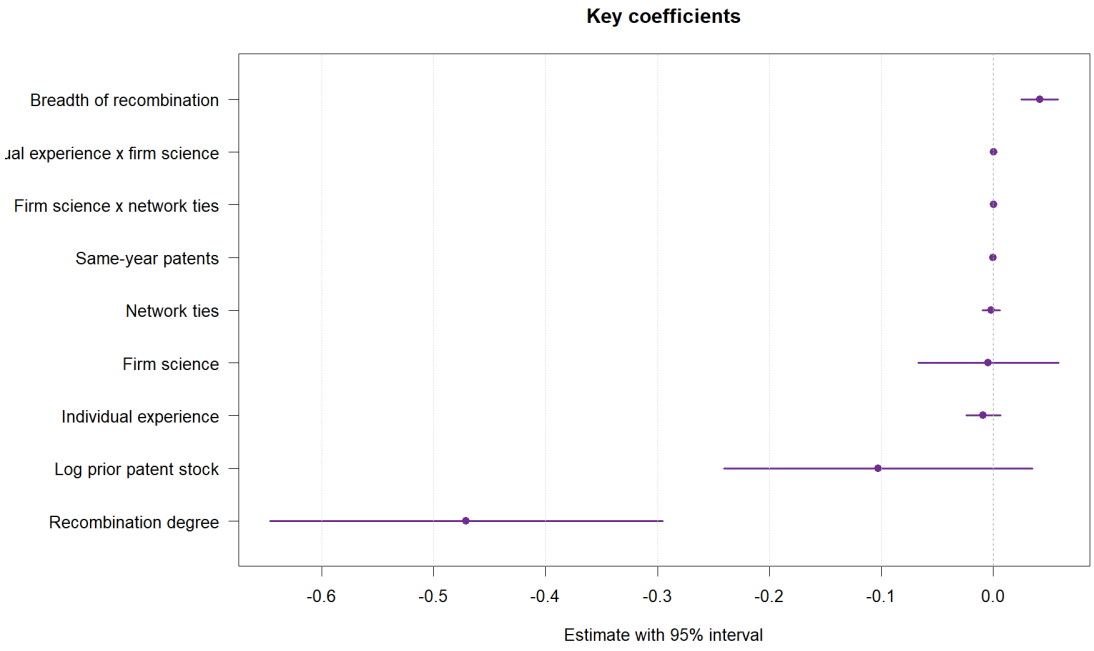

Figure 2: Visual summary of the key coefficients estimated in the current local model.

0.471 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (**???????**). The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 5.309 and standard deviation 17.53 in the local panel, while the main regressor Firm science has mean 6.772 and standard deviation 2.130 (**???????**). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.278 to 6.105 while the mean regressor ranges from 3.351 to 9.272. This does not substitute for the model, but it shows how much raw variation the specification must explain (**???????**). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The executable design is built to show whether multilevel innovation antecedents survive joint estimation. The paper therefore emphasizes not only direct effects at each level but also whether firm-level science and network ties reinforce one another. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (**???????**).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (????????). Within baseline, the leading coefficient family reads as follows: Recombination degree = -0.471 (p = ¡ .001); Breadth of recombination = 0.0413 (p = ¡ .001); Same-year patents = -4.44e-05 (p = 0.057); Log prior patent stock = -0.103 (p = 0.15); Individual experience = -0.0089 (p = 0.26). The largest positive estimate in this specification is Breadth of recombination at 0.0413, while the largest negative estimate is Recombination degree at -0.471. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (????????). This fuller coefficient reading is especially important for Rothaermel and Hess (2007) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (????????).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Recombination degree with estimate -0.471 and p = ¡ .001 (????????). The Baseline specification contributes its own reading to the replication. Recombination degree is estimated at -0.471 with p = ¡ .001 and is classified as a strong signal; Breadth of recombination is estimated at 0.0413 with p = ¡ .001 and is classified as a strong signal; Same-year patents is estimated at -4.44e-05 with p = 0.057 and is classified as a suggestive signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (????????). This model comparison matters for Rothaermel and Hess (2007) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (????????).

Term	Model	Estimate	PValue	Signal	Reading
Individual experience	Baseline	-0.0089	0.26	Weak	Negative estimate with p = 0.26.
Firm science	Baseline	-0.0047	0.89	Weak	Negative estimate with p = 0.89.
Network ties	Baseline	-0.0018	0.65	Weak	Negative estimate with p = 0.65.
Breadth of recombination	Baseline	0.0413	¡ .001	Strong	Positive estimate with p = ¡ .001.
Recombination degree	Baseline	-0.4707	¡ .001	Strong	Negative estimate with p = ¡ .001.
Same-year patents	Baseline	-4.44e-05	0.057	Suggestive	Negative estimate with p = 0.057.
Log prior patent stock	Baseline	-0.1029	0.15	Weak	Negative estimate with p = 0.15.
Individual experience x firm science	Baseline	0.0002	0.90	Weak	Positive estimate with p = 0.90.
Firm science x network ties	Baseline	2.25e-05	0.96	Weak	Positive estimate with p = 0.96.

Model-by-model interpretation table drawn from the current coefficient exports.

6.6 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (**???**). For multilevel_dynamic_capabilities, the current fit export reports N = 23,884, pseudo R-squared = 0.0504, R-squared = 0.131 (**???**). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Rothaermel and Hess (2007) is being carried out on a clearly described estimating sample with observable fit statistics (**???**).

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
multilevel_ dynamic_ capabilities	23,884	-31,504	63,419	65,084	0.0504	0.131	0.0322

Model fit statistics discussed in the main results section.

Term	Estimate	Std. error	Statistic	P-value
Individual experience	-0.0089	0.0078	-1.143	0.262
Firm science	-0.0047	0.0320	-0.146	0.885
Network ties	-0.0018	0.0039	-0.464	0.646
Breadth of recombination	0.0413	0.0083	4.978	2.48e-05
Recombination degree	-0.4707	0.0894	-5.267	1.10e-05
Same-year patents	-4.44e-05	2.24e-05	-1.984	0.0565
Log prior patent stock	-0.1029	0.0702	-1.465	0.153
Individual experience x firm science	0.0002	0.0014	0.129	0.898
Firm science x network ties	2.25e-05	0.0004	0.0538	0.957

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (**?**). For Rothaermel and Hess (2007), the local headline result is Firm Science x Network Ties = 2.25e-05 with p = 0.96. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper reports meaningful innovation antecedents at the individual, firm, and network levels, together with interaction effects across levels.	Firm Science x Network Ties = 2.25e-05 with p = 0.96.	The published paper argues that innovation is jointly shaped by antecedents at the individual, firm, and network levels. The local analogue asks whether those levels still matter jointly when mapped onto patent-level experience, firm science exposure, and citation-network ties in the current executable panel.

Aspect	OriginalPaper	LocalDataset	Reading
Measurement and data construction	The published design combines scientists, R&D capability, alliances, acquisitions, patents, publications, and citations in a multilevel pharmaceutical-biotechnology panel.	The local analogue creates a patent-level multilevel design in which patent-specific experience represents the individual layer, firm science exposure represents the firm layer, and rolling citation-network ties represent the network layer.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published sample documents the biotechnology innovation attempts of global pharmaceutical companies over 1980-2001.	Patent-level recombination panel merged with firm-year science exposure and rolling citation-network measures. Local sample = 23,901 observations across 184 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design combines scientists, R&D capability, alliances, acquisitions, patents, publications, and citations in a multilevel pharmaceutical-biotechnology panel By contrast, the local paper relies

on the following executable design: The local analogue creates a patent-level multilevel design in which patent-specific experience represents the individual layer, firm science exposure represents the firm layer, and rolling citation-network ties represent the network layer These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (**??????????**). The same caution applies to the sample itself. The published design is described as follows: The published sample documents the biotechnology innovation attempts of global pharmaceutical companies over 1980-2001 The local sample instead works from Patent-level recombination panel merged with firm-year science exposure and rolling citation-network measures and yields 23,901 observations across 184 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (**??????????**). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The published paper argues that innovation is jointly shaped by antecedents at the individual, firm, and network levels. The local analogue asks whether those levels still matter jointly when mapped onto patent-level experience, firm science exposure, and citation-network ties in the current executable panel. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (**??????????**). The comparison to the published paper is intentionally narrow. The published paper argues that innovation is jointly shaped by antecedents at the individual, firm, and network levels. The local analogue asks whether those levels still matter jointly when mapped onto patent-level experience, firm science exposure, and citation-network ties in the current executable panel. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (**????????**). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable

preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (????????).

8 Discussion

The published paper argues that innovation is jointly shaped by antecedents at the individual, firm, and network levels. The local analogue asks whether those levels still matter jointly when mapped onto patent-level experience, firm science exposure, and citation-network ties in the current executable panel (????????). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (??????). The first limitation concerns measurement. The published paper uses an unusually rich pharmaceutical-biotechnology panel with more direct multilevel observables (??????). The second limitation concerns sample translation. The local rerun uses patent-level experience, firm science exposure, and citation-network ties as transparent multilevel analogues (??????). The third limitation concerns interpretation. The replication should therefore be read as a multilevel mechanism test on the current data architecture rather than as an exact recreation of the original industry archive (??????).

9 Conclusion

The paper's contribution is a transparent multilevel analogue that can be estimated from the current patent, science, and citation-network panels (??????????). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (??????).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	23,901	rows
Unique Firms	184	unique_firms
Start Year	1986	year_min
End Year	2016	year_max
Mean Impact	5.309	mean_impact
Mean Individual Experience	1.634	mean_individual_experience
Mean Firm Science	6.772	mean_firm_science
Mean Network Ties	44.16	mean_network_ties

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	23,901	5.309	17.53	1.000	2.000	6.000	0	812
Individual experience	23,901	1.634	1.820	0.0000	1.099	2.773	0	8
Firm science	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Network ties	23,901	44.16	29.95	4.000	51.00	67.00	0	99
Breadth of recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321
Firm age	23,901	13.62	9.194	6.000	10.00	22.00	1	31

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Inventive impact	Individual experience	Firm science	Network ties	Breadth of recombination	Recombination degree	Same-year patents	Firm age
Inventive impact	1.000	-0.021	0.020	0.033	0.364	0.018	0	0.023
Individual experience	-0.021	1.000	0.044	0.032	-0.263	-0.456	0	0.125
Firm science	0.020	0.044	1.000	0.482	0.023	0.037	0	0.117
Network ties	0.033	0.032	0.482	1.000	0.094	0.067	0	0.385
Breadth of recombination	0.364	-0.263	0.023	0.094	1.000	0.466	0	0.260
Recombination degree	0.018	-0.456	0.037	0.067	0.466	1.000	0	0.156
Same-year patents	-0.018	0.060	0.463	0.338	0.011	0.065	1	0.323
Firm age	0.023	0.125	0.117	0.385	0.260	0.156	0	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	MeanImpact	MeanIndividualExperience	MeanFirmScience	MeanNetworkTies
1986	646	4.015	1.021	6.156	0.0000
1987	793	4.296	1.161	6.285	0.0000
1988	1,071	4.469	1.249	6.229	0.0000
1989	1,398	3.926	1.412	6.217	0.0000
1990	1,360	4.507	1.544	6.370	0.0000
1991	1,333	4.749	1.562	6.436	46.89
1992	1,309	5.007	1.538	6.486	53.68
1993	1,358	5.678	1.568	6.861	60.25
1994	1,337	6.272	1.596	6.866	63.28
1995	1,217	6.260	1.610	6.983	63.87
1996	659	6.727	1.802	6.880	67.66
1997	646	6.014	1.603	6.781	68.22
1998	356	6.295	1.419	6.949	68.92
1999	654	5.483	1.631	6.895	64.08
2000	636	5.541	1.686	6.781	62.11
2001	614	5.151	1.482	6.943	60.93
2002	632	4.889	1.559	6.915	59.78
2003	522	4.502	1.711	7.209	63.52
2004	324	3.664	1.479	6.939	65.01
2005	284	4.500	1.527	6.900	61.54
2006	336	4.890	1.715	7.014	60.79
2007	246	3.654	1.778	7.433	62.65
2008	535	3.652	1.712	8.930	65.82
2009	206	4.801	1.658	7.502	54.93
2010	511	4.417	1.875	7.345	53.42
2011	651	6.452	1.756	7.411	51.58
2012	1,068	4.000	1.788	7.669	51.94
2013	868	5.386	1.835	7.599	50.90

Period	Patents	MeanImpact	MeanIndividualExperience	MeanFirmScience	MeanNetworkTies
2014	725	12.64	2.050	7.636	50.58
2015	1,072	6.616	2.208	8.001	53.53
2016	534	3.768	2.707	0.0000	0.0000

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	4,783	3.351	4.747
Q2	4,809	6.196	5.349
Q3	4,754	7.163	6.095
Q4	4,837	7.904	4.278
Q5	4,718	9.272	6.105

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
individual_experience	-0.0089	0.0078	-1.143	0.26
firm_science	-0.0047	0.0320	-0.146	0.89
network_ties	-0.0018	0.0039	-0.464	0.65
breadth_recombination	0.0413	0.0083	4.978	¡ .001
recombination_degree	-0.4707	0.0894	-5.267	¡ .001
total_patents_same_year	-4.44e-05	2.24e-05	-1.984	0.057
total_patents_until_prev_log	-0.1029	0.0702	-1.465	0.15
individual_experience:firm_science	0.0002	0.0014	0.129	0.90
firm_science:network_ties	2.25e-05	0.0004	0.0538	0.96

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Inventive impact	0	0.00%
Individual experience	0	0.00%
Firm science	0	0.00%
Network ties	0	0.00%
Breadth of recombination	0	0.00%
Recombination degree	0	0.00%
Same-year patents	0	0.00%
Firm age	0	0.00%

Missingness audit for the variables summarized in the manuscript.

Entity	Label	Observations	FirstPeriod	LastPeriod
6066	INTERNATIONAL BUSINESS MACHS COR	3,200	1986	2016
5047	GENERAL ELECTRIC CO	1,318	1986	2016
4194	EASTMAN KODAK CO	1,017	1986	2010
7435	MINNESOTA MINING & MFG CO	854	1986	2016
4087	DU PONT E I DE NEMOURS & CO	830	1986	2016
4839	FORD MOTOR CO DEL	815	1986	2016
7585	MOTOROLA INC	774	1986	2016
10499	TEXAS INSTRUMENTS INC	755	1986	2016
5073	GENERAL MOTORS CORP	671	1986	2008

Entity	Label	Observations	FirstPeriod	LastPeriod
4060	DOW CHEMICAL CO	574	1986	2016
1300	ALLIED SIGNAL INC	551	1986	2016
8762	PROCTER & GAMBLE CO	524	1986	2016
1690	APPLE COMPUTER INC	452	1990	2016
1581	AMERICAN TELEPHONE & TELEG CO	421	1986	2004
10983	UNITED TECHNOLOGIES CORP	403	1986	2016
12141	MICROSOFT CORP	398	1995	2016
4503	EXXON CORP	344	1986	2016
6266	JOHNSON & JOHNSON	340	1986	2016
2285	BOEING CO	334	1986	2016
7475	MOBIL CORP	300	1986	1998

Top entities by local observation count.

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